PGA-cmdr user guide

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```
{r setup, include=FALSE} knitr::opts_chunk$set(echo = TRUE, eval = FALSE)
```

Overview

PGA-cmdr is an interactive Shiny application for exploratory and descriptive population genetics analyses from individual-based multilocus datasets (microsatellites, codominant markers). All computations run locally in R/C++ (Rcpp, OpenMP) – no data is sent to any server.

It covers the full analytical workflow:

1. Data import, with automatic loci-range and encoding detection
2. Allele frequencies per locus and population
3. General statistics – Ho, Hs, Ht, FIT/FIS/FST (Weir & Cockerham 1984), plus optional Fst-max, Fst' (Meirmans), GST/GST'' (Nei/Hedrick)
4. Null allele frequency estimation (FreeNA EM algorithm, ENA correction)
5. Local panmixia – FIS per locus and/or population (Weir & Cockerham 1984)
6. Global panmixia – FIT across all populations
7. Population subdivision – FST per locus and globally (bootstrap over individuals and loci), pairwise FST between sub-samples with CI and p-values (no GPS needed), W&C theta permutation test and a G-test
8. Genetic diversities – HS, HT (Nei & Chesser 1983), with resampling over individuals, sub-samples, or loci
9. Linkage disequilibrium – pairwise G-test with permutation p-values
10. Isolation by distance – Rousset's (1997) regression and a Mantel test

The random seed used by every bootstrap and permutation is set once, in the Import data tab.

Every module follows the same pattern: set parameters, click **Run**, and a single `.txt` (Allele Freq, Local/Global Panmixia, Subdivision, Diversities, LD) or a `.zip` of several files (General Stats, Null Alleles, IBD, Mantel) downloads automatically with everything you need – there is nothing to preview or export separately first. The exact output file name(s) are always shown just above the Run button before you click it.

Installation

PGA-cmdr is distributed through R-universe as ready-to-use binaries for Windows, macOS and Linux (no compiler needed). Install it once, or run the same command again to update to the latest version:

```
{r install} install.packages("pgacmdr",   repos = c("https://kadnaf.r-universe.dev", "https://cloud.r-project.org"))
```

Launch the application

Every time you want to use the application:

```
{r launch} pgacmdr::run_app()
```

The application opens in your default browser.

Data format

PGA-cmdr accepts CSV or tabulation-delimited TXT files. The separator (comma, semicolon, or tab) is detected automatically, and the file loads as soon as it is chosen – no extra click needed. The file must contain a header row, one *population* column, and one or more *genetic marker* columns; other columns (individual ID, latitude, longitude, etc.) are simply ignored. Two genotype encodings are supported, but a single file must use only one of them throughout.

Single-column format

Each locus occupies one column; alleles are separated by a character (`/`, `-`, `_`, etc.):

```
Population  B12      C07      D12
Pop1        192/194  145/192  92/112
Pop1        200/200  179/179  108/112
Pop2        145/145  92/92    92/100
```

Paired-column format

Each locus occupies two *adjacent* columns, one per allele. The first allele column carries the locus name (e.g. `B12`); the second must immediately follow it and be named `B12_1` (or `B12.1`):

```
Population  B12  B12_1  C07  C07_1
Pop1        192  194    145  192
Pop1        200  200    179  179
Pop2        145  145     92   92
```

Missing genotypes are coded as `0` (or `0/0`) by default and are customisable on import.

Setting the loci range on import

Once the file is loaded, a numbered **Column reference** table appears at the bottom of the Import Data page, listing every column name next to its position – use it to check exactly which number corresponds to which column.

Fill in:

1. **Population / Latitude / Longitude** – pick the matching column from each dropdown (Latitude/Longitude are optional, needed only for the IBD module's automatic geographic-distance calculation).
2. **Column number of the first locus** – where your markers start.
3. **Number of loci** – the *actual number of genetic markers* (not raw columns). As soon as the first locus column is set, this field is **auto-suggested** from how many complete loci fit between that column and the end of the file, correctly accounting for single- or paired-column format. Adjust it down if some trailing columns aren't loci.

The app shows the resulting column range live (e.g. "6 loci x 2 columns/locus = columns 9 to 20") before you click **Apply**.

Worked example: *Boophilus* cattle ticks

The package ships a demonstration dataset from Thierry de Meeus' teaching material (`inst/extdata/default_dataset.csv`): adult *Boophilus* ticks sampled from cattle across New Caledonia, genotyped at 6 microsatellite loci (B12, C07, D12, D10, A12, C03) across 8 sampling localities.

The steps below follow the app's recommended workflow (see the in-app **Help** tab for the full reference list).

1. Import the data

Click **"Load Default Data"** in the sidebar of the *Import Data* tab, or follow the manual steps above for your own file. The application imports the data via DuckDB, builds the internal genotype table (individual x locus, packed integers), and is ready for every other module immediately.

2. Allele frequencies

Set an output name on the *Allele Freq* tab (or leave it auto-filled) and click **Run**. A single `.txt` downloads with per-locus, per-population allele frequencies plus a population-size-weighted global column – every population and marker is included automatically.

3. General statistics

The *General Stats* tab reports, per locus: Ho, Hs, Ht, and FIS/FST/FIT (Weir & Cockerham 1984, in that order), plus any of the optional indices you tick (Fst-max, Fst', GST, GST''). The same Run also produces a per-population summary, the observed heterozygosity of every population at every locus, and per-allele F-statistics – all in one zip.

In the per-population summary, Ho and Hs are averaged over loci **weighted by the number of individuals typed at each locus** (loci with at least two typed individuals), which is the convention of Genepop; an unweighted mean differs from it by up to about 0.005 when loci have different amounts of missing data. The Weir & Cockerham FIS of a population is the multilocus ratio of sums.

4. Null alleles

If a locus's Ho/Hs ratio looks unusually low, check the *Null Alleles* tab: null allele frequencies are estimated per locus x sub-sample by the FreeNA EM algorithm (Chapuis & Estoup 2007), with both raw and ENA-corrected FST/DCSE reported, and a full pairwise table.

5. Local panmixia (FIS)

The *Local Panmixia* tab writes **one `.txt` file** with FIS (Weir & Cockerham 1984) **by locus** and **by population**; both are always computed. A header states the number of permutations, the type of every bootstrap and its number of replicates:

- `_indiv`: bootstrap over individuals (resampled within each sample);
- `_subs`: bootstrap over sub-samples (populations resampled as blocks), by locus. A population drawn twice counts as a distinct but identical unit and shrinks the among-population variance of the replicate; the replicates are re-centred by the bias shift (observed - mean of the replicates; Efron & Tibshirani 1993). Observed values and permutation tests are not affected;
- `_loci`: bootstrap over loci, for the Overall row and for each population.

P-values are two-sided permutation p-values. Fewer than 5 sub-samples or individuals gives `NA` rather than an unreliable interval. A second output, FIS per locus and population, gives the one-sided heterozygote deficit and excess p-values in the way FSTAT reports them.

6. Global panmixia (FIT)

The *Global Panmixia* tab writes one `.txt` file with the multilocus FIT across all populations: per-locus confidence intervals from the bootstrap over sub-samples, and two Overall rows, `Overall (BS/Ss)` (sub-samples) and `Overall (BS/Loci)` (loci), with two-sided permutation p-values.

7. Subdivision (FST)

The *Subdivision* tab writes **one `.txt` file**: (1) FST per locus (Weir & Cockerham 1984) with the bootstrap confidence interval over sub-samples and a one-sided permutation p-value based on the G statistic (as in FSTAT), followed by two Overall rows (bootstrap over sub-samples, bootstrap over loci); (2) FIS, FST, FIT, HS and HT at the multilocus level with confidence intervals from the bootstrap over loci and over sub-samples; (3) the genotype-reassignment G-test of FSTAT (Goudet et al. 1996), with its two p-value columns (`p_>=`: permuted G >= observed G; `p_>`: permuted G > observed G) and the number of complete multilocus genotypes per sample. The number of permutations of the tab is used for both tests. An option appends the bootstrap replicate values at the end of the file so that their distributions can be inspected.

8. Genetic diversities (HS, HT)

The *Diversities* tab writes one file, `Diversities_confidence_intervals_<date>.txt`, with HS (within-population) and HT (total) gene diversity – using the **unbiased small-sample estimator of Nei & Chesser (1983)**, not the simpler biased form: HS per locus (bootstrap over individuals and over sub-samples), HS per population (over individuals and over loci), HT per locus, and the multilocus estimators (over loci and over sub-samples).

Resampling individuals makes every replicate slightly less diverse than the sample (the bootstrap mean is about HS/(2n) below the observed HS), so a plain percentile interval hugs the observed value. The intervals from the bootstrap over individuals are therefore shifted by (observed HS - bootstrap mean); the bootstrap over sub-samples shows no such bias and is left as is.

9. Linkage disequilibrium

The *LD* tab writes one `.txt` file. It tests every locus pair, per population and combined (“All”), using a permutation-based log-likelihood-ratio (G) test. One locus’s genotypes are permuted among individuals within each population (10,000 permutations by default) and compared to the observed G statistic. Loci, locus pairs and populations are listed in the order of the data file.

```
{r ld-note, eval=FALSE} # G = 2 * sum(observed * log(observed / expected)), under independence
```

10. Isolation by distance

The *IBD* tab offers two independent analyses:

- **Rousset’s (1997) regression** of linearised genetic distance (FST/(1-FST)) against geographic distance – this module **reuses the pairwise table already computed by the Null Alleles module** rather than recomputing it, so run Null Alleles first (or upload an external pairwise file). Geographic distance is computed automatically from Latitude/Longitude set at import.
- **A Mantel permutation test** (Pearson, Spearman, and Rousset 1D/2D statistics), independent of the regression above.

Statistical methods

All F-statistics follow **Weir & Cockerham (1984)**:

> Weir, B.S. & Cockerham, C.C. (1984). Estimating F-statistics for the analysis of population structure. *Evolution*, **38**, 1358–1370.

Gene diversity (HS, HT) follows the unbiased small-sample estimator of **Nei & Chesser (1983)**:

> Nei, M. & Chesser, R.K. (1983). Estimation of fixation indices and gene diversities. *Annals of Human Genetics*, **47**, 253–259.

Null allele frequencies and the ENA correction follow **Chapuis & Estoup (2007)** (the FreeNA method):

> Chapuis, M.P. & Estoup, A. (2007). Microsatellite null alleles and estimation of population differentiation. *Molecular Biology and Evolution*, **24**, 621–631.

Isolation by distance follows **Rousset (1997)**:

> Rousset, F. (1997). Genetic differentiation and estimation of gene flow from F-statistics under isolation by distance. *Genetics*, **145**, 1219–1228.

The genotype-reassignment G-test for subdivision follows **FSTAT**:

> Goudet, J., Raymond, M., de Meeus, T. & Rousset, F. (1996). Testing differentiation in diploid populations. *Genetics*, **144**, 1933–1940.

Fst-max and Fst′ follow **Meirmans (2006)** and **Hedrick (2005)**; GST′′ follows **Meirmans & Hedrick (2011)**:

> Meirmans, P.G. & Hedrick, P.W. (2011). Assessing population structure: FST and related measures. *Molecular Ecology Resources*, **11**, 5–18.

Permutation p-values use the formula `p = (b + 1) / (m + 1)`, where `b` is the number of permuted statistics at least as extreme as the observed value and `m` is the number of permutations, following:

> Phipson, B. & Smyth, G.K. (2010). Permutation p-values should never be zero: calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, **9**, 39.

Heavy computations (bootstrap, permutation loops) are parallelised with OpenMP via C++17 (Rcpp).

Confidence intervals: the three bootstraps

PGA-cmdr offers three non-parametric bootstraps. They do not measure the same source of variability, so their intervals are not expected to be equal.

BS/Indiv (bootstrap over individuals). Individuals are resampled with replacement *within* each sub-sample; the sub-samples and their sizes are kept. This is the default for FIS, FIT, FST and HT (per locus and over all loci, with the multilocus ratio of sums recomputed on each replicate) and for HS. It measures the sampling error due to the individuals actually typed.

BS/Loci (bootstrap over loci). Loci are resampled with replacement and the multilocus estimators are recomputed from the per-locus variance components. It measures the variability between loci (used for the multilocus values only).

BS/Ss (bootstrap over sub-samples). Whole sub-samples are resampled with replacement. It is kept only for HS per locus, FIS per locus and in Null Alleles (as in FreeNA). It is not used any more for FST, FIT and HT.

Why BS/Ss was replaced for FST, FIT and HT

When r sub-samples are drawn with replacement among r, some are drawn twice and are treated as two distinct but identical units. If s^2 is the variance among sub-samples of the allele frequencies (divisor r-1), the average among-sub-sample variance of a replicate is only $s^2(r-1)/r$. FST, FIT and HT are functions of this among-sub-sample variance, so their replicates were centred too low (on a simulated island model with 8 sub-samples of 50 individuals, only about 0.7% of the replicates exceeded the observed FST and the 95% interval excluded the estimate). FIS and HS do not depend on this variance and were not affected.

For the remaining BS/Ss uses (HS and FIS per locus, Null Alleles) the replicates are re-centred with the published bias shift (Efron & Tibshirani 1993, ch. 10): each replicate is replaced by replicate - mean(replicates) + observed value, so that the interval is the percentile interval moved by (observed - mean of the replicates). Its width is not changed. Observed values and permutation tests are not modified. This replaces an earlier, unpublished rescaling of the among-sub-sample variance by r/(r-1), which was abandoned in favour of the published method.

Even after the bias shift, BS/Ss intervals for FST remain wider than the sampling variability when the number of sub-samples is small. In a simulation of an island model (FST about 0.012, 10 loci, 50 individuals per sub-sample, 40 simulated metapopulations per case, 300 bootstrap replicates), the mean bootstrap standard error of the overall FST was:

	True SD of FST			

Number of sub-samples	(new sample of sub-samples)	BS/Ss	BS/Indiv	BS/Loci
5	0.0013	0.0030	0.0016	0.0012
10	0.0009	0.0013	0.0011	0.0009
20	0.0007	0.0007	0.0008	0.0006
40	0.0004	0.0004	0.0006	0.0004

BS/Ss is therefore unreliable for FST with few sub-samples (about three times too large with 5), which is why it is no longer used for FST, FIT and HT.

Bias shift of the BS/Indiv intervals

Resampling individuals duplicates genotypes, so the replicates are slightly biased. In the simulations (island model, 8 sub-samples of 50 individuals, 10 loci, 40 simulated data sets) the mean of the bootstrap replicates minus the observed value was:

Statistic	HS	FST	FIS	FIT	HT
Bootstrap bias (BS/Indiv)	-0.0092	+0.0099	-0.0101	0.0000	-0.0011
SD of the estimator	0.0024	0.0009	0.0046	0.0046	0.0023

Duplicated genotypes lower the within-sub-sample diversity (HS) and, mechanically, raise FST; FIS is lowered by about 1/(2n) for a sub-sample of n individuals; the two effects cancel in FIT = 1 - (1 - FIS)(1 - FST), which is unbiased; HT is almost unaffected. Each replicate R is therefore corrected by the estimated bias, R(corrected) = R + (observed - mean of the replicates), which amounts to shifting the percentile interval by (observed - bootstrap mean) (bias correction of the bootstrap, Efron & Tibshirani 1993, *An Introduction to the Bootstrap*, chapter 10). The correction is applied to every statistic and every level (per locus, per sub-sample, overall); the width of the interval and the standard error (SE, the standard deviation of the replicates) are not modified; the permutation p-values are not affected. For FIT the correction is negligible.

For the bootstrap over sub-samples the replicates of HT are lowered (a sub-sample drawn twice reduces the among-sub-sample variance), and so are those of FST and FIT, whereas HS is not affected (simulation, 8 sub-samples: HT -0.0023, FST -0.0027, FIT -0.0027, HS 0.0000). The same shift, observed - mean of the replicates, is applied to these BS/Ss intervals.

A note on cross-module dependencies

Most modules are fully self-contained: they read the imported dataset and compute everything themselves. The one exception is **Isolation by Distance**, whose Rousset regression reuses the Null Alleles module's pairwise table instead of recomputing it. If you go directly to IBD without having run Null Alleles first, the app will tell you so and ask you to run it (or upload an external pairwise file instead). Keep this in mind if you change your imported data or parameters between the two – re-run Null Alleles before trusting IBD's regression output.

Session info

```
{r session} sessionInfo()
```